Lecture 1: Introduction and Dynamic Modelling

第一讲：导论与动态建模

Computational Methods for Heterogeneous-Agent Macro
异质性主体宏观的计算方法

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Welcome / 欢迎

Goal of This Course
本课程目标

Solving heterogeneous-agent GE models with aggregate shocks in Julia.

在 Julia 中求解带总量冲击的异质性主体一般均衡模型。

First half: Standard (with some suggestive notational decisions)
Second half: My own research

前半部分：标准内容（含一些有启发性的记号选择）
后半部分：我自己的研究

Every solution will be based on the **Bellman equation**.

所有求解都基于**贝尔曼方程**。

Heterogeneous Agents: Motivation 1
异质性主体：动机一

Empirically, average marginal propensity to consume (MPC) is high, but average savings is also high.
A representative agent would not act like this. So a representative-agent model cannot explain this.
However, if poor households have high MPC/low saving, and rich households have low MPC/high saving, then a **heterogeneous-agent** model can explain.

经验上，平均边际消费倾向 (MPC) 很高，但平均储蓄也很高。
代表性主体不会这样行动。所以代表性主体模型无法解释这一现象。
但若贫穷家庭具有高 MPC、低储蓄，富裕家庭具有低 MPC、高储蓄，则**异质性主体**模型可以解释。

Heterogeneous Agents: Motivation 2
异质性主体：动机二

A change in (e.g. interest rate) policy will affect different people **differently**.
A policy will have different **aggregate** effects depending on who it targets.
We care about the distributional and aggregate consequences of policy.

一项（例如利率）政策的变化会对不同的人产生**不同的**影响。
一项政策的**总量**效应取决于它针对的是谁。
我们关心政策的分布效应和总量后果。

Schedule / 日程

	English	**中文**
Lecture 1	Introduction & dynamic modelling	导论与动态建模
Lecture 2	Julia. Value function iteration	Julia. 价值函数迭代
Lecture 3	Heterogeneity beyond wealth	财富之外的异质性
Lecture 4	Equilibrium. Aiyagari	均衡。Aiyagari
Lecture 5	Aiyagari transitions	Aiyagari 过渡动态
Lecture 6	Uncertainty. Krusell–Smith	不确定性。Krusell–Smith
Lecture 7	**CVIAYN**	**续值即一切**
Lecture 8	Stage decomposition	阶段分解
Lecture 9	Discrete choice. Spatial	离散选择。空间模型
Lecture 10	Applications. Estimation	应用。估计

Practicalities
实务

- Each lecture has a paired Jupyter notebook.
- Please stop me if you have any questions!

- 每讲都配有一个 Jupyter 笔记本。
- 有问题请随时打断我！

Household Problem / 家庭问题

Microfoundation: One Household
微观基础：单一家庭

Macroeconomics is the study of how individual behaviour shapes aggregate outcomes.
To understand the origins of macro outcomes, need to study the sources of individual behaviour.

宏观经济学研究的是个体行为如何塑造总量结果。
要理解宏观结果的来源，就需要研究个体行为的成因。

Two Period Model
两期模型

Household starts with **wealth** b_0 in Period 0. In Period 1, they receive income w_1.

家庭在第 0 期持有**财富** b_0。第 1 期获得收入 w_1。

$$\max_{c_0,c_1\geq 0} \; u(c_0)+\beta u(c_1) \quad \text{s.t.} \quad c_0+\frac{c_1}{R}\leq \overline{W}.$$

Where $\overline{W}=b_0+w_1/R$ is **lifetime resources**.

其中 $\overline{W}=b_0+w_1/R$ 是**终生资源**。

If you save \$1 today, you can spend \$$R$ tomorrow, but utility discounted by β.

若今天储蓄 \$1，明天可消费 \$$R$，但效用按 β 折现。

Lagrangian:
$\mathcal{L}=u(c_0)+\beta u(c_1)-\lambda(c_0+c_1/R-\overline{W})$.

拉格朗日量：
$\mathcal{L}=u(c_0)+\beta u(c_1)-\lambda(c_0+c_1/R-\overline{W})$。

Euler Equation
欧拉方程

Lagrangian FOCs: $u'(c_0) = \lambda$, $\beta u'(c_1) = \lambda / R$. Rearranging,

拉格朗日一阶条件：$u'(c_0) = \lambda$，$\beta u'(c_1) = \lambda / R$。整理得：

$$\boxed{\frac{u'(c_0)}{u'(c_1)} = \beta R.}$$

At optimum, on margin, **indifferent** between consuming or saving one unit today for R units tomorrow.

在最优处，边际上对今天消费一单位、或储蓄换取明天 R 单位**无差异**。

Example: Analytical Solution with Log Utility
例：对数效用下的解析解

$$u = \log c \;\Rightarrow\; c_1 = \beta R c_0 \;\Rightarrow$$

$$c_0^\star = \frac{\overline{W}}{1+\beta}, \qquad c_1^\star = \frac{\beta R \overline{W}}{1+\beta}.$$

With $\beta = 0.96,\ R = 1.04,\ b_0 = 10,\ w_1 = 2$:

$$c_0^\star \approx 6.08, \quad c_1^\star \approx 6.07.$$

Bumpy income $(10, 2) \;\Rightarrow\;$ smooth consumption $(6.08, 6.07)$.

取 $\beta = 0.96$，$R = 1.04$，$b_0 = 10$，$w_1 = 2$：

$$c_0^\star \approx 6.08, \quad c_1^\star \approx 6.07.$$

不平的收入 $(10, 2) \;\Rightarrow\;$ 平滑的消费 $(6.08, 6.07)$。

Policy Function
策略函数

The **Policy Function** represents optimal period-0 consumption in terms of **initial wealth** b_0:

策略函数用**初始财富** b_0 表示第 0 期的最优消费：

$$c_0^\star(b_0) = \frac{b_0 + w_1/R}{1+\beta}.$$

This way, we have a model of household behaviour, and **how** household behaviour **arises**.

这样我们就得到了一个家庭行为的模型，以及家庭行为**是如何产生**的。

Preferences
偏好

Standard utility:

$$u(c) = \frac{c^{1-\gamma} - 1}{1-\gamma}, \quad \gamma > 0.$$

Utility **increasing** in c:

$$\frac{\partial u(c)}{\partial c} = c^{-\gamma} > 0.$$

Marginal utility **decreasing**:

$$\frac{\partial^2 u(c)}{\partial c^2} = -\gamma c^{-\gamma-1} < 0.$$

标准效用函数：

$$u(c) = \frac{c^{1-\gamma} - 1}{1-\gamma}, \quad \gamma > 0.$$

效用随 c **递增**：

$$\frac{\partial u(c)}{\partial c} = c^{-\gamma} > 0.$$

边际效用**递减**：

$$\frac{\partial^2 u(c)}{\partial c^2} = -\gamma c^{-\gamma-1} < 0.$$

T Period Model
T 期模型

$$\max_{c_0,\ldots,c_{T-1}} \sum_{t=0}^{T-1} \beta^t u(c_t), \qquad b_{t+1} = R(b_t - c_t) + w_{t+1}.$$

Solving Lagrangian gives us $T-1$ Euler equations,

求解拉格朗日量给出 $T-1$ 个欧拉方程：

$$u'(c_t) = \beta R\, u'(c_{t+1}).$$

This is already 80 equations in 80 unknowns!

这已经是 80 个方程、80 个未知量！

Stochastic Income
随机收入

Suppose $\{w_t\}$ follows a Markov chain with transition matrix Π.

Given today's w_t, we only know the *distribution* of w_{t+1}, not the value.

At every time t, household chooses c_t without knowing w_{t+1}.

So what's the policy function? c_t is a function of t variables! That is, $c_t(w_0, \dots, w_t)$.

设 $\{w_t\}$ 服从转移矩阵为 Π 的马尔可夫链。

给定今天的 w_t，我们只知道 w_{t+1} 的分布，而不知道具体取值。

在每个时刻 t，家庭在不知道 w_{t+1} 的情况下选择 c_t。

那么策略函数是什么？c_t 是 t 个变量的函数！即 $c_t(w_0, \dots, w_t)$。

Stochastic Euler Equation 随机欧拉方程

With $T = 2$, substitute / 取 $T = 2$ 代入：

$$c_1 = R(b_0 - c_0) + w_1 .$$

Then / 于是：

$$\max_{c_0} \; u(c_0) + \beta\, \mathbb{E}[\, u(R(b_0 - c_0) + w_1)\,] .$$

FOC with respect to c_0 / 对 c_0 的一阶条件：

$$\boxed{u'(c_0) \;=\; \beta R\, \mathbb{E}[\, u'(c_1)\,] .}$$

The Euler equation now carries an expectation: saves against an *uncertain* c_1.

欧拉方程现在带期望：对不确定的 c_1 储蓄对冲。

Infinite-Horizon Problem ($T = \infty$)
无限期问题 ($T = \infty$)

With $T = \infty$, the household / 当 $T = \infty$ 时，家庭：

$$\max \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t), \quad \text{s.t.} \quad b_{t+1} = R(b_t - c_t) + w_{t+1}.$$

Specifically, must find the best **policy functions** / 具体地，需找到最优**策略函数**：

$$c_t = c_t(w_0, \dots, w_t) \quad \forall t.$$

There are infinitely-many possible decision rules, so we need to approach the problem differently.

可能的决策规则有无穷多种，所以我们需要换一种思路。

Recursive Form / 递归形式

Richard Bellman's First Observation
Bellman 的第一个观察

At each time t, household only makes one choice, then waits,

在每个时刻 t，家庭只做一个选择，然后等待：

$$\max_{c_t} = u(c_t) + \beta \mathbb{E}_t \left[\max \mathbb{E}_{t+1} \sum_{s=t+1}^{\infty} \beta^t u(c_s) \right]$$

$$s.t. \quad b_{s+1} = R(b_s - c_s) + w_{s+1} \quad \forall s \geq t.$$

Importantly, the problem at time t looks almost identical to the problem at time $t+1$.
The only possible differences are b_t and w_t.
These are the **state variables.**

关键是：t 时刻的问题与 $t+1$ 时刻的问题几乎一模一样。
唯一可能的差别是 b_t 和 w_t。它们就是**状态变量**。

State Variables
状态变量

In a recursive model, **state variables** are the only variables that make time periods (and decisions) **different from each other.** Here, the state is (b, w).

在递归模型中，**状态变量**是唯一使得各时期（及其决策）**相互不同**的变量。
这里，状态是 (b, w)。

Richard Bellman's Second Observation
Bellman 的第二个观察

Richard Bellman realized two things:

1. You can express his insight as an equation about a **value function** V.
2. If you know the **value function**, you can compute the **policy function** and (numerically) solve the entire model.

Bellman 意识到两件事：

1. 他的洞见可以表达为关于**价值函数** V 的一个方程。
2. 若知道**价值函数**，就可以计算出**策略函数**，并（数值地）求解整个模型。

Value Function and Bellman Equation
价值函数与贝尔曼方程

Define the *value function* $V(b, w)$ as the **net present value** of state (b, w). / 定义价值函数 $V(b, w)$ 为状态 (b, w) 的**净现值:**

$$V(b_t, w_t) = \max_{c_t \in [0, b_t]} \mathbb{E}_t \sum_{s=t}^{\infty} \beta^{s-t} u(c_s)$$
$$\text{s.t.} \quad b_{s+1} = R(b_s - c_s) + w_{s+1} \forall s.$$

Then V satisfies the **Bellman equation** / 则 V 满足**贝尔曼方程:**

$$\boxed{V(b, w) = \max_{c \in [0, b]} u(c) + \beta \, \mathbb{E}\left[V(b', w') \,\middle|\, w \right]}$$

where $b' = R(b - c) + w'$ / 其中 $b' = R(b - c) + w'$。

The Value Function Is All You Need
价值函数即一切

If we know V, then we can compute the **policy function** and solve the whole model,

若我们知道 V，就可以计算**策略函数**并求解整个模型：

$$c^\star(b,w) = \arg\max_{c\in[0,b]} u(c) + \beta\,\mathbb{E}[\,V(R(b-c)+w,\,w')\mid w\,]\,.$$

How do we find V?
如何求 V?

We still have a problem, though. We don't just need to find a number. We need to find an entire **function** V.
Idea: Think about what would happen if we got the value function **wrong**.

不过我们还有一个问题。我们要找的不只是一个数，而是一个完整的**函数** V。
思路：想一想如果把价值函数**猜错**了会怎样。

The Bellman operator on arbitrary v
作用在任意 v 上的贝尔曼算子

Suppose that we guess a **function** v. Even if we got that function **wrong**, this would still predict some policy,

设我们猜一个**函数** v。即便**猜错**，它仍然会给出某个策略：

$$c(b, w; v) = \arg\max_{c\in[0,b]} u(c) + \beta\,\mathbb{E}[\,v(R(b-c)+w',\, w') \mid w\,]\,.$$

If we use v to **look ahead**, we can get a **different function** $\mathcal{T}v$,

若用 v 去**前瞻**，可以得到一个**不同的函数** $\mathcal{T}v$：

$$(\mathcal{T}v)(b, w) \;:=\; \max_{c\in[0,b]} u(c) + \beta\,\mathbb{E}\Big[\,v(R(b-c)+w',\, w')\,\Big|\,w\Big]\,.$$

The Lookahead Operator
前瞻算子

Idea: Think of $\mathcal{T}$ as an **operator** that maps a **function** to a **function**: $v \mapsto \mathcal{T}v$.
This is the **Lookahead Operator**

Read $\mathcal{T}v$ as: "if v were the value-from-tomorrow-on, this is what the value-today would be."

思路：把 $\mathcal{T}$ 看作把**函数**映到**函数**的**算子**：$v \mapsto \mathcal{T}v$。
这就是**前瞻算子**（Lookahead Operator）。

$\mathcal{T}v$ 读作："若 v 代表明天起的价值，那么今天的价值就是这个"。

V is the fixed point of $\mathcal{T}$
V 是 $\mathcal{T}$ 的不动点

The Bellman relation, restated:

将贝尔曼关系重述为：

$$\boxed{V = \mathcal{T}V.}$$

The true value function is a fixed point of $\mathcal{T}$.

So solving the household's problem reduces to: *find a fixed point of* $\mathcal{T}$.

真实的价值函数是 $\mathcal{T}$ 的一个不动点。

于是求解家庭问题就归结为：求 $\mathcal{T}$ 的一个不动点。

Existence and uniqueness
存在性与唯一性

Under mild conditions, $\mathcal{T}$ is a contraction:

在温和的条件下，$\mathcal{T}$ 是一个压缩映射：

$$\|\mathcal{T}v_1 - \mathcal{T}v_2\|_\infty \leq \beta \|v_1 - v_2\|_\infty .$$

Banach: a unique fixed point V^* exists.

Iterating $v_{k+1} = \mathcal{T}v_k$ from any starting v_0 converges to V^* at rate β.

That's also next week's algorithm.

Banach 不动点定理：存在唯一的不动点 V^*。

从任意 v_0 出发迭代 $v_{k+1} = \mathcal{T}v_k$，以速率 β 收敛到 V^*。

这也是下周的算法。

One equation to take home
带走的一条方程

$$\boxed{V(b,w) \;=\; \max_{c\in[0,b]} \; u(c) + \beta\,\mathbb{E}\big[\, V(R(b-c)+w',\, w') \,\big|\, w \,\big]\,.}$$

State: (b, w). Unknown: V.

Everything in the rest of the course is a variation on this.

状态：(b, w)。未知量：V。

本课程余下的一切，都是它的变体。

From one household to a population / 从单一家庭到总体分布

Where we are
目前所在位置

- One household.
- Prices given.
- State (b, w).
- Value V, policy $c^\star$.

Five extensions left. I'll preview each.

- 一个家庭。
- 价格给定。
- 状态 (b, w)。
- 价值 V，策略 $c^\star$。

还剩五个扩展，下面逐一预览。

Extension 1a: many households (lecture 3)
扩展 1a：多个家庭（第三讲）

Same primitives, different states (b_i, w_i).

A single $c^\star(b, w)$ tells you what *every* household does.

The first hint of "CVIAYN": one function describes optimal behavior for everyone.

New object: distribution $\Lambda(b, w)$.

原语相同，状态 (b_i, w_i) 不同。

一个 $c^\star(b, w)$ 就告诉你每一个家庭的行为。

"CVIAYN" 的第一个线索：一个函数描述所有人的最优行为。

新对象：分布 $\Lambda(b, w)$。

Extension 1b: aggregates from the distribution
扩展 1b：从分布得到总量

$$\bar{K} = \int b \, d\Lambda, \qquad \bar{C} = \int c^{\star}(b, w) \, d\Lambda.$$

Push Λ_t through $c^{\star}$ and the income shock to get Λ_{t+1}.

In a stationary world, $\Lambda_t \to \Lambda^{\star}$. Lecture 3 sets it up.

用 $c^{\star}$ 和收入冲击把 Λ_t 推到 Λ_{t+1}。

在平稳情形下，$\Lambda_t \to \Lambda^{\star}$。第三讲给出这套设定。

Extension 2: general equilibrium (lecture 4)
扩展 2：一般均衡（第四讲）

R is no longer given. Firms pay $R = 1 + F_K(\bar{K}, L) - \delta$.

But $\bar{K} = \int b \, d\Lambda$, Λ depends on $c^\star$, $c^\star$ depends on R:

$$R \to c^\star \to \Lambda \to \bar{K} \to R.$$

A fixed point. This is Aiyagari.

R 不再给定。企业按 $R = 1 + F_K(\bar{K}, L) - \delta$ 支付。

但 $\bar{K} = \int b \, d\Lambda$，$\Lambda$ 依赖 $c^\star$，$c^\star$ 依赖 R:

$$R \to c^\star \to \Lambda \to \bar{K} \to R.$$

一个不动点。这就是 Aiyagari。

Extension 3: time-varying prices (lecture 5)
扩展 3：时变价格（第五讲）

$$V_t(b, w) = \max_c u(c) + \beta \mathbb{E}[V_{t+1}(R_t(b-c)+w', w') \mid w].$$

After a permanent policy change, R_t moves over time. Households forecast $\{R_t\}$.

Backward sweep on V_t, forward on Λ_t, outer fixed point on $\{R_t\}$.

永久性政策变化之后，R_t 随时间变化。家庭预测 $\{R_t\}$。

对 V_t 倒推，对 Λ_t 正推，对 $\{R_t\}$ 求外层不动点。

Extension 4: aggregate shocks (lecture 6, K-S)
扩展 4：总量冲击（第六讲，K–S）

$\{R_t\}$ is now driven by aggregate shocks Z_t (productivity, etc.).

R_{t+1} depends on $\bar{K}_{t+1}$, which depends on tomorrow's Λ, which depends on today's.

So Λ_t is part of the household's state:

$$(b_t, w_t, \Lambda_t, Z_t).$$

Infinite-dimensional.

现在 $\{R_t\}$ 由总量冲击 Z_t（生产率等）驱动。

R_{t+1} 依赖 $\bar{K}_{t+1}$，依赖明天的 Λ，依赖今天的 Λ。

于是 Λ_t 也是家庭状态的一部分：

$$(b_t, w_t, \Lambda_t, Z_t).$$

无穷维。

The Krusell–Smith trick
Krusell–Smith 技巧

Approximate Λ by a few moments — often just $\bar{K}$.

Postulate a law of motion for $\bar{K}$, solve, simulate, regress, update, iterate.

Works remarkably well ("approximate aggregation"). But it's a workaround.

用几个矩近似 Λ——往往就用 $\bar{K}$。

假设 $\bar{K}$ 的运动律，求解、模拟、回归、更新、迭代。

效果出奇地好（"近似聚合"）。但毕竟只是一种权宜之计。

Extension 5: CVIAYN (lecture 7)
扩展 5：CVIAYN（第七讲）

$$V_\theta(b, w, \mathbf{m}(\Lambda), Z) \text{ as a neural network.}$$

Sample states. Compute Bellman residual. Push the squared residual down by SGD.

The slogan: V encodes everything about the future. Learn V well and you've solved the model.

采样状态。计算贝尔曼残差。用 SGD 把平方残差压下去。

口号：V 编码了关于未来的一切。把 V 学好就等于求解了模型。

The arc, with lecture numbers
课程脉络（带课次）

One household, given prices / 单一家庭，价格给定 (lecture 1, today / 第一讲，今天)

↓

Many households, exogenous prices / 多家庭，外生价格 (lecture 3 / 第三讲)

↓

Stationary GE: Aiyagari / 稳态一般均衡：Aiyagari (lecture 4 / 第四讲)

↓

Aiyagari transitions / Aiyagari 过渡 (lecture 5 / 第五讲)

↓

Krusell–Smith (lecture 6 / 第六讲)

↓

Continuation Value is All You Need / 续值即一切 (lecture 7 / 第七讲)

↓

Stages, discrete choice, spatial / 阶段、离散选择、空间 (lectures 8–10 / 第八–十讲)

One equation, ten lectures
一个方程，十讲

$$V(\text{ state }) = \max_{\text{action}} \text{ flow payoff} + \beta \mathbb{E}[V(\text{ next state })].$$

$$V(\text{ 状态 }) = \max_{\text{动作}} \text{ 当期收益} + \beta \mathbb{E}[V(\text{ 下期状态 })].$$

- L2: solve numerically.
- L3: rich state.
- L4: R endogenous.
- L5: time-varying.
- L6: Λ in state.
- L7: $V =$ neural net.
- L8–10: stages, discrete choice, applications.

- L2：数值求解。
- L3：丰富的状态。
- L4：R 内生。
- L5：时变。
- L6：Λ 进入状态。
- L7：$V =$ 神经网络。
- L8–10：阶段、离散选择、应用。

How we’ll work / 工作方式

Why Julia
为什么用 Julia

- Fast (within 2$\times$ of C).
- Math-like syntax. `β = 0.96` is valid.
- Multiple dispatch.
- Increasingly used in macro / SciML.

- 速度快（与 C 相差 2$\times$ 以内）。
- 数学化语法。`β = 0.96` 是合法的。
- 多重分派。
- 在宏观 / SciML 中使用日益增多。

Notebooks
笔记本

Each lecture has a Jupyter notebook in `tutorial_notebooks/`.

Today's is `L01_intro.ipynb`: state, transition, payoff, Bellman operator as a Julia function. No solving yet — lecture 2.

每讲在 `tutorial_notebooks/` 下都有一个 Jupyter 笔记本。

今天是 `L01_intro.ipynb`：状态、转移、收益、贝尔曼算子作为一个 Julia 函数。今天还不求解——那是第二讲的事。

Setup checklist
安装清单

1. Julia 1.10+
   `https://julialang.org/downloads/`
2. VS Code + Julia extension.
3. `using Pkg; Pkg.add("IJulia")`.
4. Jupyter (Anaconda or `pip install jupyter`).

1. Julia 1.10+
   `https://julialang.org/downloads/`
2. VS Code + Julia 扩展。
3. `using Pkg; Pkg.add("IJulia")`。
4. Jupyter（Anaconda 或 `pip install jupyter`）。